

SMIT ROY | JAVA BACKEND DEVELOPER

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PROFESSIONAL SUMMARY

Java Backend Developer with hands-on experience building production-grade REST APIs using Spring Boot, Spring Data JPA, and Spring Security. Skilled in clean layered architecture, transactional workflows, and real-world integrations including payment gateways and JWT-based auth. Strong focus on clean architecture, scalable design, and maintainable code.

TECHNICAL SKILLS

Languages: Java

Frameworks: Spring Boot, Spring Data JPA, Spring Security, Spring Boot Actuator

Databases: PostgreSQL, MySQL, Redis

Cloud & Deploy: Docker, Render, Neon Serverless, GitHub Actions (CI/CD)

Dev Tools: Maven, Git/GitHub, Postman, DBaver, JUnit, Mockito, Swagger

Concepts: OOP, JWT & OAuth2, Basic System Design, REST API Design, Concurrency Control

OPEN SOURCE

JWT STARTER DEPENDENCY FOR SPRING BOOT | [jwt-spring-boot-starter](#) - [GitHub \(1.0.2\)](#)

May 2026

- Built a reusable JWT authentication starter for Spring Boot with auto-configuration and zero boilerplate setup
- Implemented **JwtService**, **JwtAuthenticationFilter**, and **SecurityFilterChain** as auto-configured beans with **@ConditionalOnMissingBean** for full extensibility
- Published to **GitHub Packages** with Maven and an automated CI/CD pipeline using **GitHub Actions**
- Designed plug-and-play architecture allowing developers to override any component via custom bean definitions

MAJOR PROJECTS

STAYGRID | Hotel Rental Booking Backend - [GitHub](#)

March 2026 - Present

- Designed and developed a scalable backend system for **property listings** and **bookings** using Spring Boot
- Implemented **JWT-based authentication** with refresh tokens and role-based access control (GUEST / HOTEL_MANAGER)
- Developed booking logic with **date conflict resolution** and **transactional workflows**, preventing overlapping reservations across 365 days of forward inventory per room
- Integrated **pagination** and **filtering** (location, price range) to optimize API performance; persisted data with PostgreSQL deployed on **Neon Serverless Postgres**
- Applied **DTO pattern**, **global exception handling**, and standardized API responses for a maintainable and consistent architecture
- Integrated **Redis caching** for **hotel search results** to reduce database load on read-heavy endpoints; used **@CacheEvict** to maintain cache consistency on hotel updates
- Implemented **dynamic hotel pricing** using the **Strategy Pattern** with layered surge, occupancy, and urgency-based adjustments, running on an hourly scheduler
- Integrated **Stripe payment gateway** with webhook-based confirmation and automated refund handling for cancelled bookings
- Designed a multi-state inventory management system (reserved → confirmed → cancelled) to maintain booking consistency
- Implemented concurrency control using pessimistic locking to prevent overbooking under multi-user concurrent scenarios

CLINIXHUB | Hospital Management System - [GitHub](#)

April 2026 - May 2026

- Built a production-grade Hospital Management REST API covering patient management, doctor onboarding, and appointment booking with a clean layered architecture
- Implemented stateless **JWT authentication** with a custom **OncePerRequestFilter** and integrated **OAuth2 social login (Google)** with automatic user provisioning
- Designed a **hybrid Role + Permission-based access control system** (ADMIN, DOCTOR, PATIENT) using Spring Security 6's **@PreAuthorize** and **@Secured** at both URL and method level
- Modelled complex JPA relationships (**@ManyToOne**, **@OneToMany**, **@ManyToMany**) and wrote **custom JPQL queries** for aggregation, pagination, and bulk updates
- Built a **centralized exception handling layer** using **@RestControllerAdvice**, returning consistent **ApiError** responses for auth, JWT, and access denied failures

LMS PORTAL | Learning Management System - [Render](#) | [GitHub](#) | [YouTube](#)

January 2026 - March 2026

- Built a RESTful backend API using Spring Boot to manage **students**, **courses**, **instructors**, and **enrollments**
- Designed relational data models with JPA (**One-to-Many**, **Many-to-One** mappings) for a normalized schema
- Developed a course enrollment system with duplicate prevention logic and enrollment tracking
- Implemented input validation and global exception handling to ensure robust and predictable API behavior
- Created a **preloaded dataset (data.sql)** to simulate real-world LMS scenarios for testing and demonstration
- Containerized the application with **Docker** and deployed it on **Render** with PostgreSQL integration
- Documented and demonstrated the full system end-to-end via a public YouTube walkthrough

PROFESSIONAL EXPERIENCE

TASK VIRTUAL SERVICES

Associate

Kolkata, India

March 2024 - Present

- Developed and maintained 15+ client websites across real estate, manufacturing, education, and e-commerce, handling end-to-end development and delivery
- Built and customised WordPress plugins using PHP hooks and filters to extend core functionality per client requirements
- Wrote custom PHP for theme functionality, business logic, and server-side behaviour across multiple client projects
- Implemented custom UI components and page-level logic using HTML, CSS, JavaScript, and page builders like Elementor, tailored to individual client requirements
- Integrated third-party APIs and payment gateways, and plugin configurations; connected marketing workflows
- Managed MySQL databases directly via phpMyAdmin, handled schema changes, data migrations, and query-level debugging across live client projects
- Managed hosting configuration, domain setup, deployment pipelines, and CMS workflows, ensuring smooth go-live for each project

Tech Stack: HTML, CSS, JavaScript, PHP, MySQL, WordPress

EDUCATION

CHANDIGARH UNIVERSITY

Master of Computer Applications

Specialization - **Cloud Computing**

Current Semester GPA: 9.0

Punjab, India

July 2025 - August 2027

UNIVERSITY OF CALCUTTA

Bachelor's Degree

Cumulative GPA: 7.0

Kolkata, India

July 2021 - August 2024